

VenuMadhavi

Data Analyst

Phone: (314)657-7983 | Email: [reddymadhavi163@gmail.com](mailto:rednymadhavi163@gmail.com) | Location: Saint Louis, MO 63108 [LinkedIn](#)

SUMMARY

- Highly motivated and analytical **Data Analyst** with **4+ years** of progressive experience transforming raw, complex data into actionable business insights that improve decision-making and drive measurable outcomes.
- Skilled in **SQL, Python, Power BI, Tableau, dbt**, and **Snowflake**, with a strong foundation in data wrangling, statistical modeling, data pipeline automation, and data visualization.
- Proven track record of delivering business-critical dashboards and reports to executive leadership while reducing reporting bottlenecks and increasing operational efficiency.
- Possesses hands-on experience in building robust ETL workflows with Airflow and dbt, designing cloud-based analytical platforms on **AWS** and **GCP**, and conducting A/B testing, time-series forecasting, and customer segmentation to solve real-world business problems.
- Adept at cross-functional collaboration with stakeholders, product teams, and engineers to deliver data solutions that align with organizational KPIs.
- A strong communicator who thrives in **Agile** environments, passionate about building scalable data products and driving a culture of data-driven decision-making.

EDUCATION

Master’s in computer and information science Saint Louis University – Saint Louis, MO	Aug 2023 – May 2025
Bachelor of Computer Science and Engineering Sathyabama Institute of Science and Technology – Chennai, India	June 2015 – April 2019

SKILLS

Languages: Python, SQL, R, DAX, Bash
Libraries & ML: Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Statsmodels, XGBoost, NLTK, Ggplot2
Data Tools: Tableau, Power BI, Looker, Excel, Ggplot2
Databases: Snowflake, PostgreSQL, SQL Server, MySQL, MongoDB
Cloud Platforms: AWS (S3, RDS, Redshift), GCP (BigQuery, Cloud Storage)
Data Engineering Tools: dbt, Apache Airflow, ETL/ELT, REST APIs
Other Tools: Jupyter Notebook, VS Code, Git, Docker (basic), MS Office Suite
Practices: A/B Testing, Forecasting, Data Governance, CRISP-DM, Agile/Scrum

EXPERIENCE

Sept 2024 Present	UnitedHealth Group, MO Data Analyst	- Designed and automated ETL pipelines using Apache Airflow and dbt to streamline ingestion and transformation of healthcare claims, eligibility, and provider datasets, reducing reporting latency by 40% across three business units. - Migrated legacy BI workflows to a modern Snowflake-based cloud data warehouse, enabling faster query performance (65% improvement) for actuarial, operations, and risk management teams. - Developed over 20 Power BI and Tableau dashboards to monitor KPIs such as claim denial rates, readmission trends, member engagement, and provider network utilization, enhancing real-time visibility for senior leadership.
------------------------------	--	---

- Utilized Crystal Reports to create, maintain, and optimize healthcare and operational reporting, improving report delivery speed and accuracy for stakeholders.
- Managed data provisioning processes, ensuring timely and secure delivery of healthcare datasets to internal teams, third-party vendors, and public health agencies.
- Collaborated with public health agencies and internal compliance teams, aligning data analysis with population health initiatives and CMS regulatory guidelines.
- Performed member segmentation using K-means clustering on 1.2M+ enrollee records, identifying high-risk cohorts and enabling personalized care outreach campaigns, increasing engagement ROI by 19%.
- Leveraged LLMs (e.g., GPT-4) to automate SQL query generation, Python code review, and documentation tasks, resulting in a 25% increase in productivity for ad hoc analytics and compliance reporting.
- Integrated OpenAI APIs to develop proof-of-concept chatbots for internal data access (e.g., benefit utilization queries) and to auto-generate pipeline documentation for internal audits.
- Applied Responsible AI practices by auditing bias in risk stratification models and ensuring compliance with CMS guidelines for data fairness and transparency.
- Conducted A/B testing on digital member portal optimizations (e.g., appointment reminders, wellness nudges), using T-tests and Chi-square analysis to boost conversion rates by 11%.
- Built time-series forecasting models (ARIMA, Prophet, XGBoost) for predicting claims volume, ER visits, and plan utilization, improving resource planning accuracy by 18%.
- Collaborated with cross-functional teams including actuaries, underwriters, care coordinators, and engineering to translate healthcare business needs into data solutions aligned with quality metrics and regulatory reporting.
- Established robust data governance practices by documenting data dictionaries, building metadata frameworks, and leading initiatives that improved onboarding and interdepartmental data literacy.

Aug 2019
March 2023

Zaven IT Solutions Pvt. Ltd, Hyderabad India
Data Engineer

- Engineered scalable data infrastructure for processing large, complex datasets, aligning with both functional and non-functional business requirements.
- Designed and implemented internal process improvements, including re-architecting infrastructure for enhanced scalability, automating data workflows, and improving operational efficiency.
- Developed end-to-end ETL pipelines using AWS (S3, Redshift, Glue) and SQL-based data transformations, enabling efficient data extraction, transformation, and loading from diverse sources.
- Built robust analytical tools and AI-powered pipelines to deliver actionable business insights, focusing on customer behavior analysis, performance metrics, and predictive modeling.
- Collaborated with cross-functional teams (Data Science, Product, Design, and Executive teams) to translate data into strategic decisions and resolve complex technical issues.
- Led initiatives to integrate AI/ML-driven data solutions into existing systems, boosting automation and reducing manual interventions across business operations.
- Championed a data-driven culture by mentoring teams on best practices for leveraging cloud-native tools and AI-enhanced analytics platforms.